05 – Frequency Response of Single-stage Amplifiers

Winter Term 2025/26

Overview

- **Control Theory & System Characterization – A Summary**
- Two-port Model of the MOS Transistor
- CS Amplifier – Frequency Response
- Miller-Effect

RC Lowpass Filter – Time versus Frequency Domain

Kirchhoff loop $v_\mathrm{I} = v_\mathrm{O} + Ri$

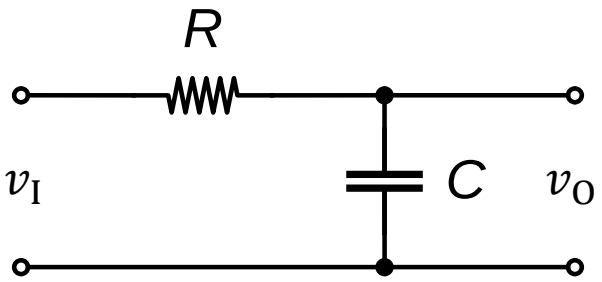
capacitor $i = C\dfrac{\partial v_\mathrm{O}}{\partial t}$

Kirchhoff loop $v_\mathrm{I} = v_\mathrm{O} + RC\dfrac{\partial v_\mathrm{O}}{\partial t}$

Laplace domain $v_\mathrm{I} = v_\mathrm{O} + sRCv_\mathrm{O}$

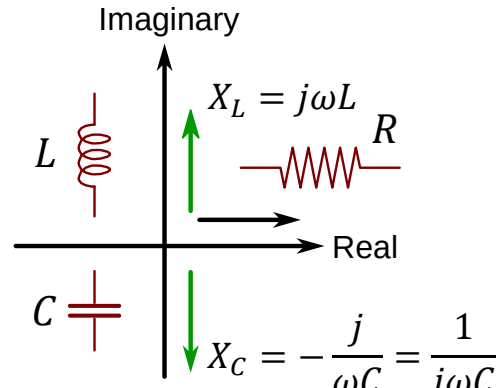
transfer function $H(s) = \dfrac{v_\mathrm{O}(s)}{v_\mathrm{I}(s)} = \dfrac{1}{1+sRC}$

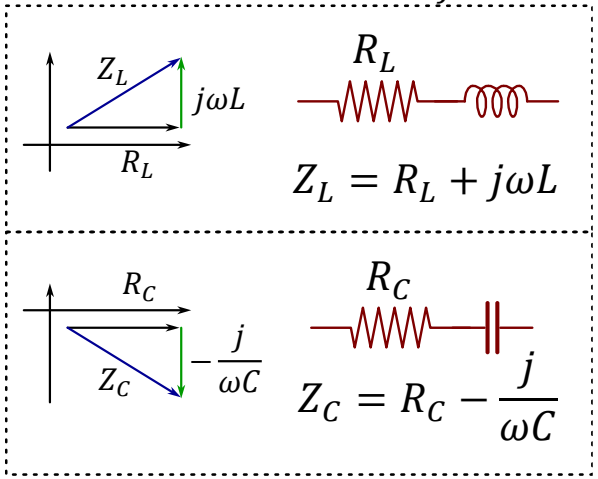
frequency response $H(j\omega) = \dfrac{v_\mathrm{O}(j\omega)}{v_\mathrm{I}(j\omega)} = \dfrac{1}{1+j\omega RC}$



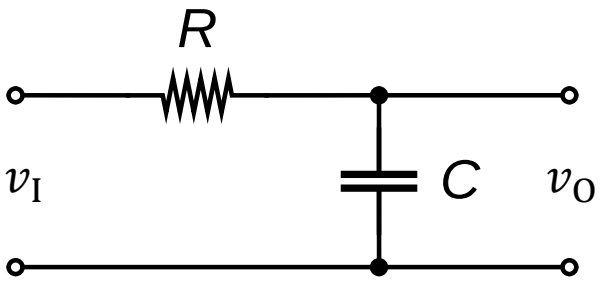
$$\frac{\partial}{\partial t} \rightarrow s, \qquad \int \partial t \rightarrow \frac{1}{s}$$

$$s = j\omega = j2\pi f$$

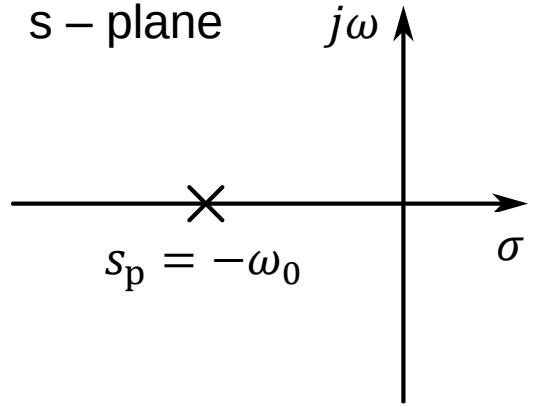




RC Lowpass Filter – Transfer Function



single real pole
characteristic frequency of the circuit

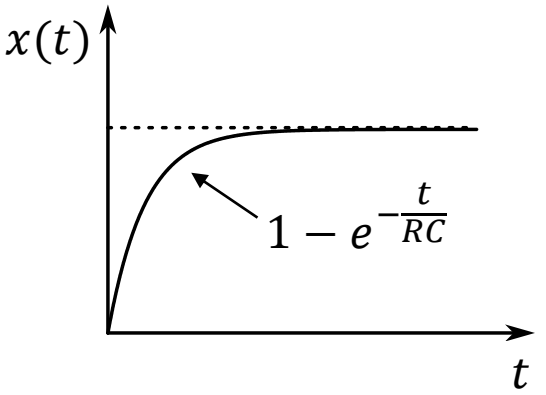


transfer function $\dfrac{v_\mathrm{O}}{v_\mathrm{I}} = \dfrac{1}{1+sRC}$

$$s_\mathrm{p} = \sigma_\mathrm{p} = -\frac{1}{RC} = -\omega_0 = -2\pi f$$

step response $x(t)\Big|_{1(t)} = 1 - e^{-\frac{t}{RC}}$

time constant $\tau = RC$



A single negative real pole yields an exponentially settling behavior.

The closer the pole to the origin and thus to the imaginary axis, i.e., the larger the time-constant, the longer the settling!

RC Lowpass Filter – Frequency Response & Bode Plot

Bode plot = graphical representation of the amplitude and phase of $H(j\omega)$ for positive ω

example: RC low-pass filter

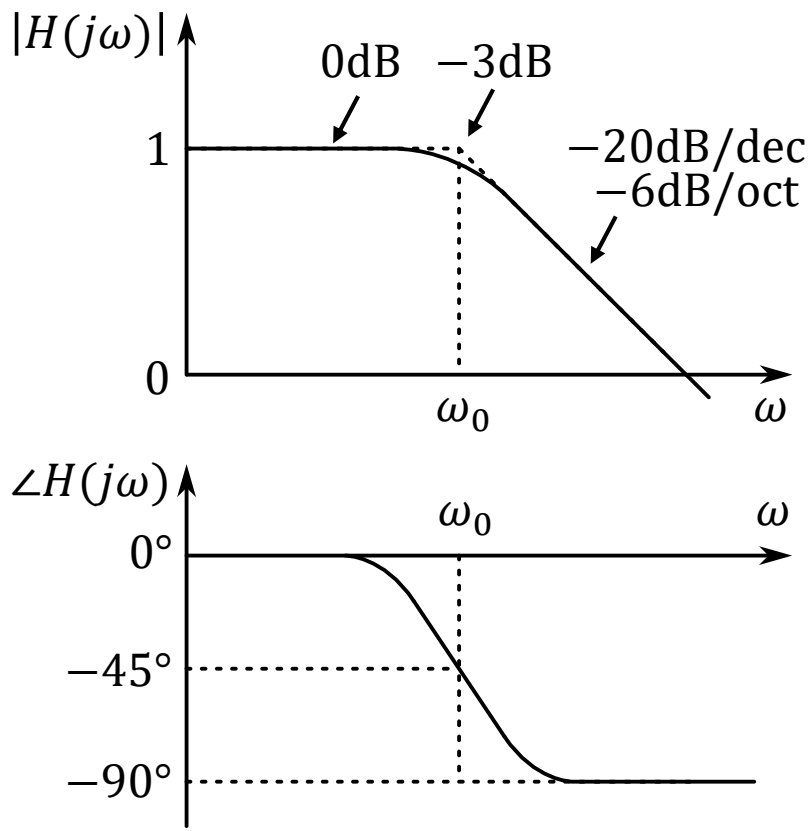
$$H(s) = \frac{1}{1+sRC} \rightarrow H(j\omega) = \frac{1}{1+j\omega RC} = \frac{1-j\omega RC}{1+(\omega RC)^2}$$

$$|H(j\omega)| = \frac{1}{|1+j\omega RC|} = \frac{1}{\sqrt{1+(\omega RC)^2}}$$

$$= \underbrace{1}_{\omega = 0} \quad / \quad \underbrace{1/\sqrt{2}}_{\omega = 1/RC} \quad / \quad \underbrace{0}_{\omega \rightarrow \infty}$$

$$\phi_{H(j\omega)} = \arctan\left(\frac{\mathrm{Im}\{H(j\omega)\}}{\mathrm{Re}\{H(j\omega)\}}\right) = \arctan(-\omega RC)$$

$$= \underbrace{0\,^\circ}_{\omega = 0} \quad / \quad \underbrace{-45\,^\circ}_{\omega = 1/RC} \quad / \quad \underbrace{-90^\circ}_{\omega \rightarrow \infty}$$



CR Highpass Filter – Frequency Response & Bode Plot

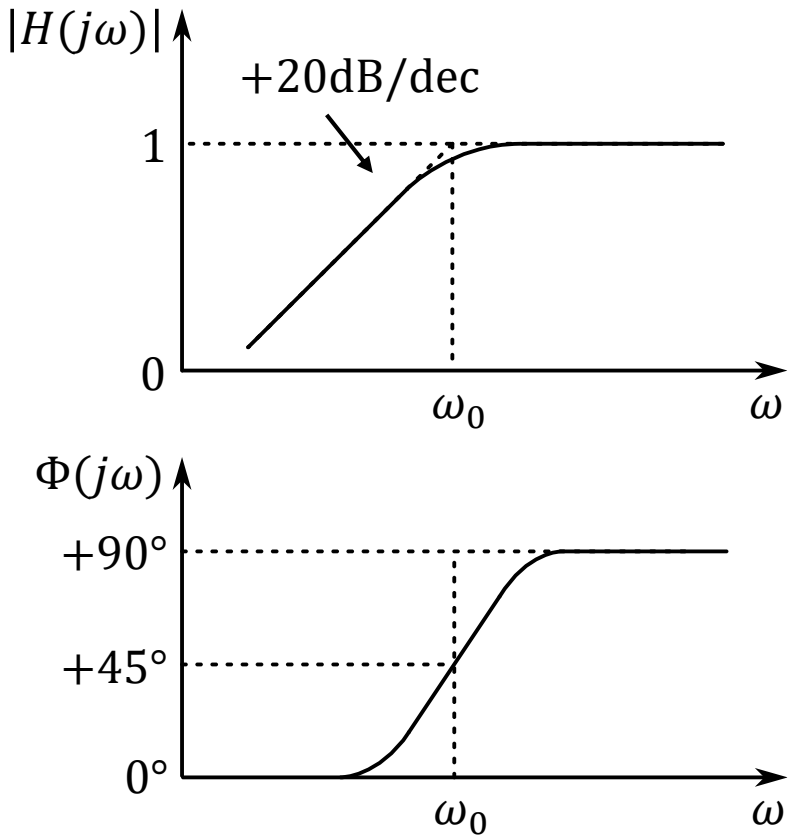
$$H(s) = \frac{sRC}{1+sRC} \rightarrow H(j\omega) = \frac{j\omega RC}{1+j\omega \mathrm{RC}}$$

$$|H(j\omega)| = \frac{\omega RC}{\sqrt{1+(\omega RC)^2}}$$

$$= \underbrace{0}_{\omega = 0} \quad / \quad \underbrace{1/\sqrt{2}}_{\omega = 1/\mathrm{RC}} \quad / \quad \underbrace{1}_{\omega \rightarrow \infty}$$

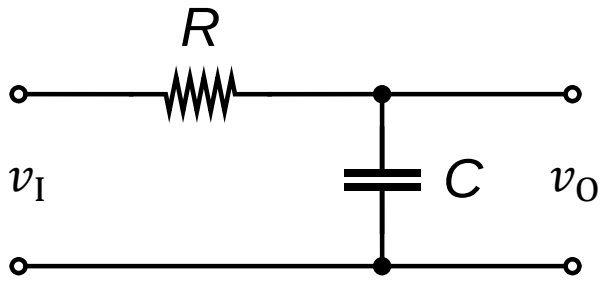
$$\Phi(j\omega) = \arctan\left(\frac{Im\{H(j\omega)\}}{Re\{H(j\omega)\}}\right) = \arctan(+\omega RC)$$

$$= \underbrace{0\,°}_{\omega = 0} \quad / \quad \underbrace{45\,°}_{\omega = 1/\mathrm{RC}} \quad / \quad \underbrace{90°}_{\omega \rightarrow \infty}$$



Bode Plot – Summary I

- Each negative pole ω_p (located on the negative real axis in the left half of the s-plane) of the transfer function
  - causes a -90° phase shift (rule of thumb: starts at $\omega_p/10$, completes at $10\omega_p$)
  - decreases the magnitude by -20dB / decade
- At $\omega = \omega_p$
  - phase shift: -45°
  - magnitude: -3dB → -3dB corner or cut-off frequency



- Each negative zero ω_z (located on the negative real axis in the left half of the s-plane) of the transfer function
  - causes a +90° phase shift (rule of thumb: starts at $\omega_z/10$, completes at $10\omega_z$)
  - increases the magnitude by +20dB / decade
- At $\omega = \omega_z$
  - phase shift: +45°
  - magnitude: -3dB

Frequency Response – Poles & Zeros

Rewrite the transfer function from polynomial to product form:

$$H(s) = \frac{b_0 + b_1 s + \cdots + b_m s^m}{1 + a_1 s + \cdots + a_n s^n} = \frac{k \prod_{i=1}^{m}(1 + \frac{s}{z_i})}{\prod_{i=1}^{m}(1 + \frac{s}{p_i})}$$

z_i → zero

p_i → pole

Poles and zeros are either real or complex-conjugate.

magnitude

$$|H(s)| = |k| \frac{\sqrt{1 + \left|\frac{s}{z_1}\right|^2}\sqrt{1 + \left|\frac{s}{z_2}\right|^2}\cdots}{\sqrt{1 + \left|\frac{s}{p_1}\right|^2}\sqrt{1 + \left|\frac{s}{p_2}\right|^2}\cdots}$$

$$20 \cdot \log|H(s)| = 20 \cdot \log|k| + 20 \cdot \log\sqrt{1 + \left|\frac{s}{z_1}\right|^2} + 20 \cdot \log\sqrt{1 + \left|\frac{s}{z_2}\right|^2} + \cdots$$
$$-20 \cdot \log\sqrt{1 + \left|\frac{s}{p_1}\right|^2} - 20 \cdot \log\sqrt{1 + \left|\frac{s}{p_2}\right|^2} - \cdots$$

Independent of sign!
pos./neg. zero → +20dB/decade
pos./neg. pole → -20dB/decade

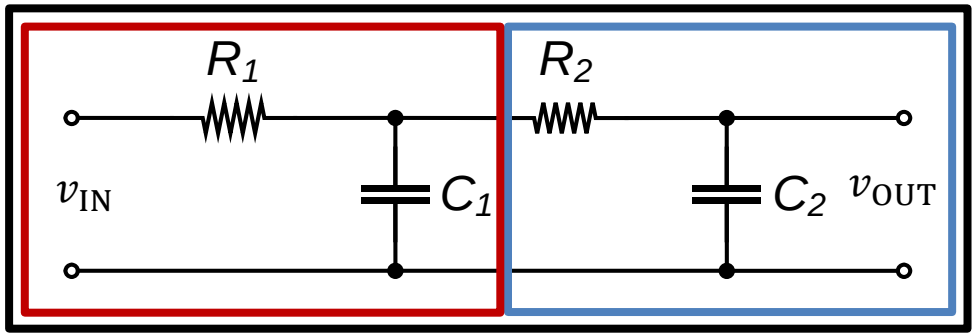
phase

$$\Phi = \arctan\left(\frac{Im\{H(s)\}}{Re\{H(s)\}}\right)$$
$$= \arctan\left(\frac{\omega}{z_1}\right) + \arctan\left(\frac{\omega}{z_2}\right) + \cdots$$
$$- \arctan\left(\frac{\omega}{p_1}\right) - \arctan\left(\frac{\omega}{p_2}\right) - \cdots$$

Depending on sign!
pos./neg. zero → -/+ 90°
pos./neg. pole → +/- 90°

Example – 2nd-order Low-pass Filter

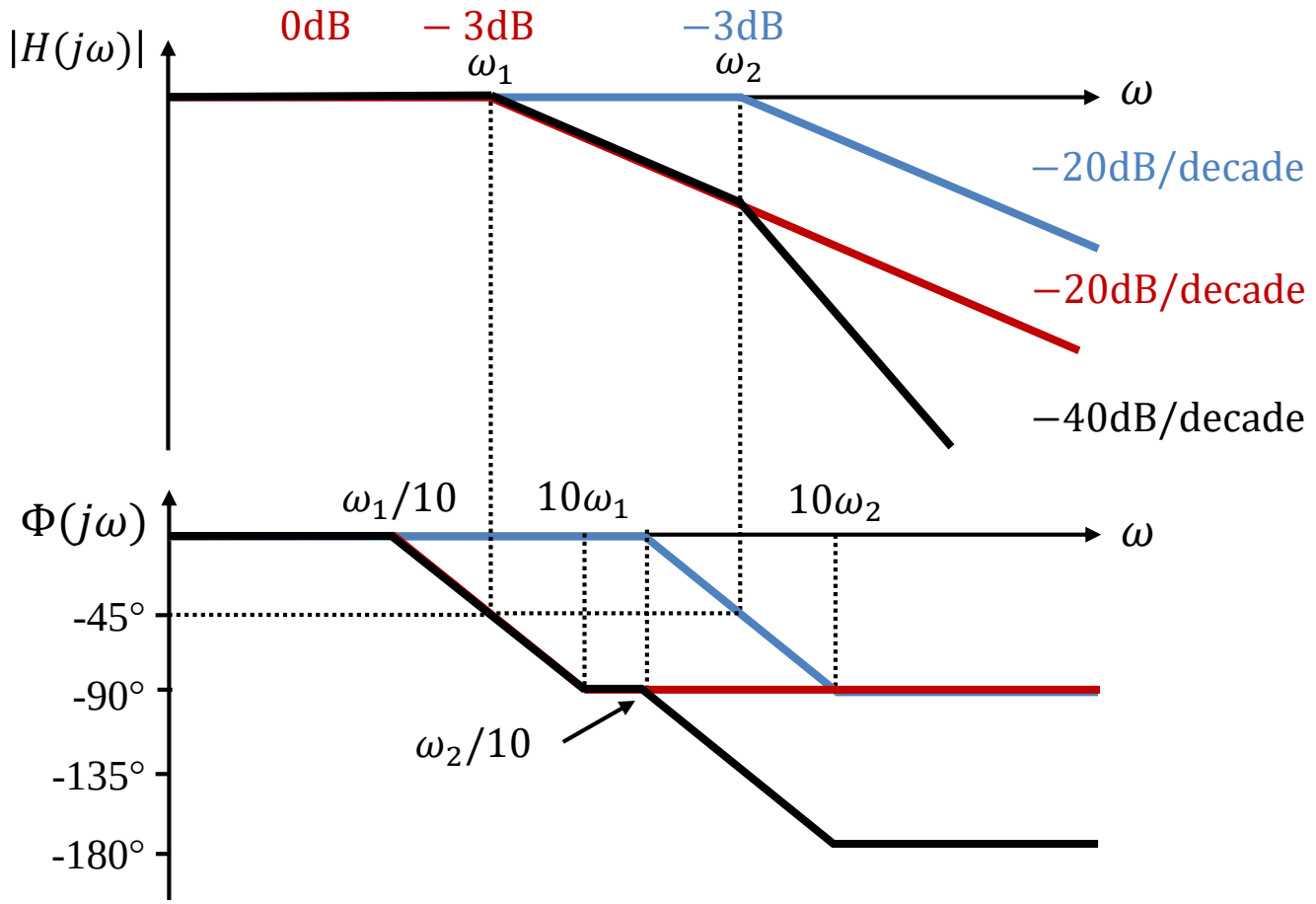
1st-order low-pass 1st-order low-pass



2nd-order low-pass $H(s)$

$$H(s) = \frac{1}{1 + s/p_1} \frac{1}{1 + s/p_2}$$

with $p_2 > 10p_1 \leftrightarrow \omega_2 > 10\omega_1$



Bode Plot – Summary II

- Overall gain of systems connected in series

  → product of the individual gains of each system

- Overall phase shift of systems connected in series

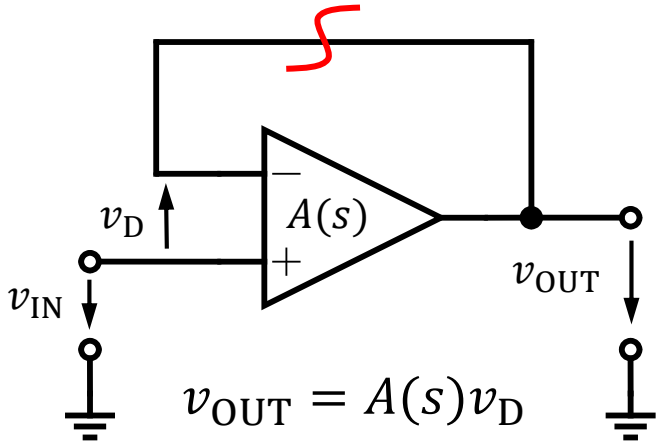
  → sum of the individual phase shifts of each system

- Instability occurs if a system exhibits a positive pole ω_p or a complex-conjugate pole with positive real part (located in the right half s-plane).

- Positive zeroes (located on the positive real axis in the right half of the s-plane) are acceptable.
- Each positive zero ω_z
  - causes a -90° phase shift (rule of thumb: starts at $\omega_z/10$, completes at $10\omega_z$)
  - increases the magnitude by $+20\text{dB/decade}$

Stability of Feedback Systems I

What is the closed-loop transfer function v_{OUT}/v_{IN} of this circuit?



$$v_{IN} = v_D + v_{OUT} = \left(\frac{1}{A(s)} + 1\right) v_{OUT}$$

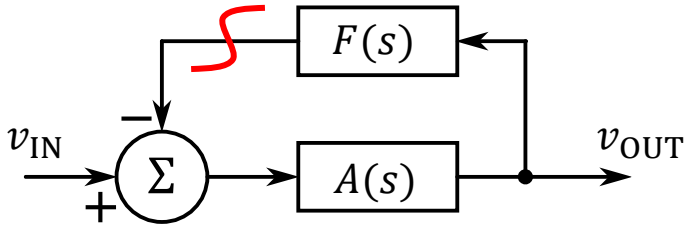
$$\frac{v_{OUT}}{v_{IN}} = \frac{A(s)}{1 + A(s)} = \frac{1}{1 + \frac{1}{A(s)}}$$

open-loop transfer function

$$A(s) \approx A_0 \frac{1}{1 + \frac{s}{\omega_{p1}}} \frac{1}{1 + \frac{s}{\omega_{p2}}}$$

Frequency response of the loop transfer function $H_0(j\omega)$ gives insight into the stability & small-signal settling behavior of the CLOSED-LOOP transfer function.

general case

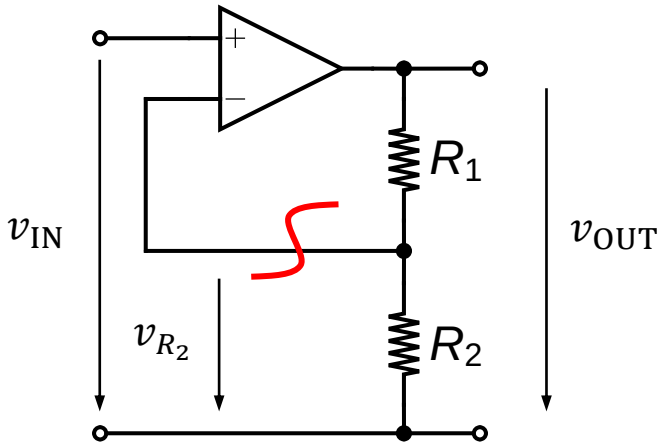


$$\frac{v_{OUT}}{v_{IN}} = \frac{A(s)}{1 + F(s)A(s)}$$

closed-loop transfer function

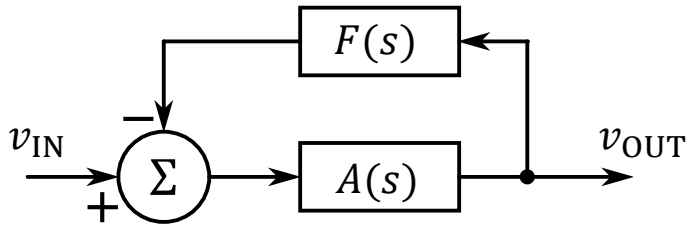
$$H_0(s) = A(s)F(s)$$

(open-) loop transfer function



Stability of Feedback Systems II

general case



$$\frac{v_{\mathrm{OUT}}}{v_{\mathrm{IN}}} = \frac{A(s)}{1 + H_0(s)}$$ closed-loop transfer function

$$H_0(s) = A(s)F(s)$$ loop transfer function

$1 + H_0(s)$ characteristic polynomial

$1 + H_0(s) = 0$ characteristic equation

The solutions $s_{\mathrm{p}} = \sigma_{\mathrm{p}} \pm j\omega_{\mathrm{p}}$ (either real or complex-conjugate) of the characteristic equation

are the poles of the closed-loop transfer function

determine the homogenous intrinsic response of the system (without external excitation)

$$v_{out}(t) = c_1 e^{s_{\mathrm{p}1}t} + c_2 e^{s_{\mathrm{p}2}t} + \cdots + c_{\mathrm{n}} e^{s_{\mathrm{pn}}t}$$

and thus the stability of the system, i.e.,

$\sigma_{\mathrm{p}} < 0 \rightarrow$ stable system $\leftrightarrow \sigma_{\mathrm{p}} > 0 \rightarrow$ instable system

How to Analyze the Stability?

Yes/no statement on stability of a closed-loop system!

- stable: all poles s_p are located in the negative s-plane ($\sigma_\mathrm{p} < 0$)
- instable: one or more poles s_p are located in the positive s-plane ($\sigma_\mathrm{p} > 0$)

mathematical approach

- determine the poles by solving the characteristic equation $1 + \mathrm{H}_0(s) = 0$

graphical approach

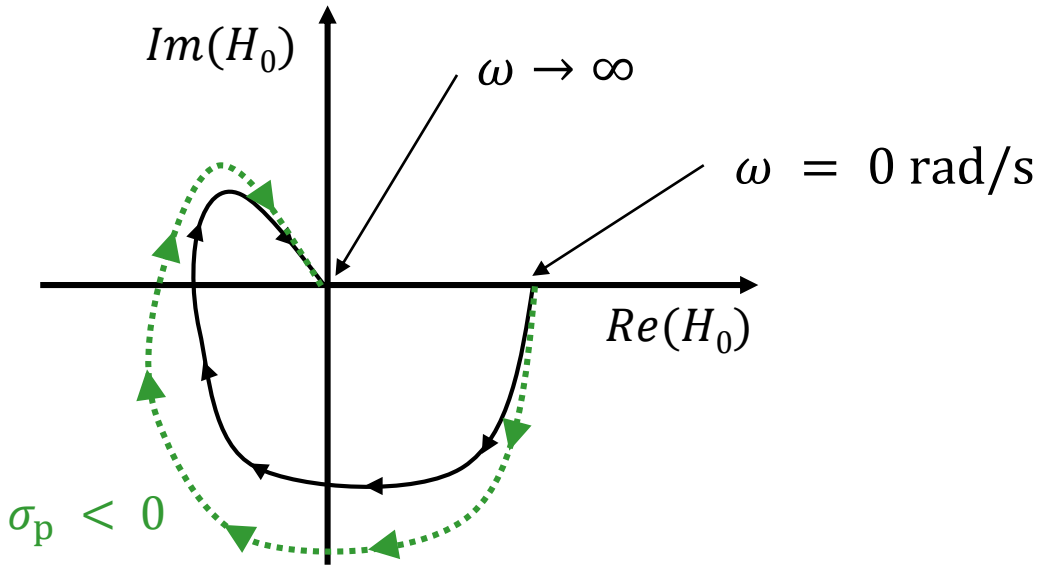
- Nyquist plot
- Bode plot

Nyquist Plot

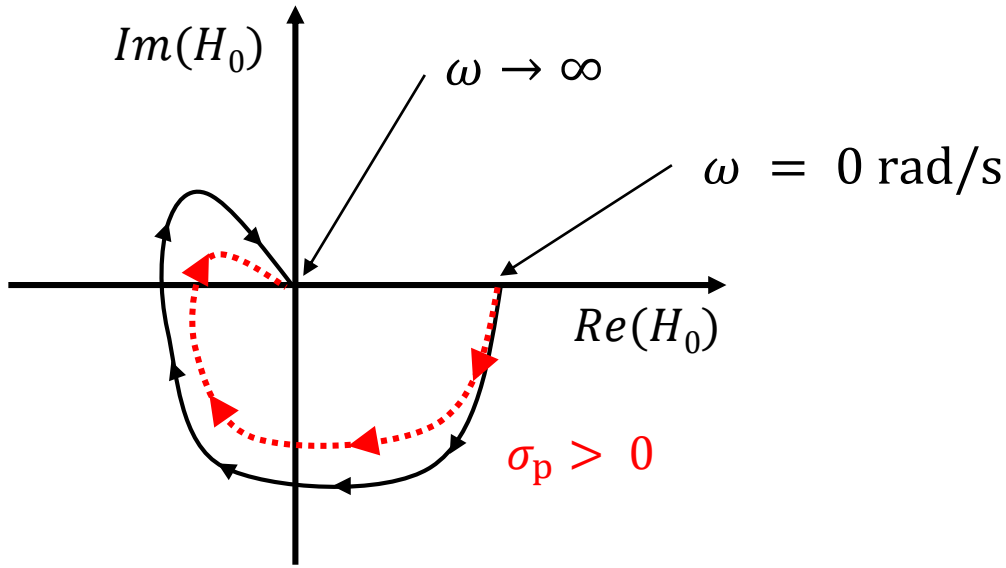
Nyquist plot

=

graphical representation of the loop transfer function $H_0(j\omega)$, i.e., $\sigma_\mathrm{p} = 0$



Nyquist curves with $\sigma_\mathrm{p} < 0$ are located on the left-hand side of the Nyquist curve with $\sigma_\mathrm{p} = 0$



Nyquist curves with $\sigma_\mathrm{p} > 0$ are located on the right-hand side of the Nyquist curve with $\sigma_\mathrm{p} = 0$

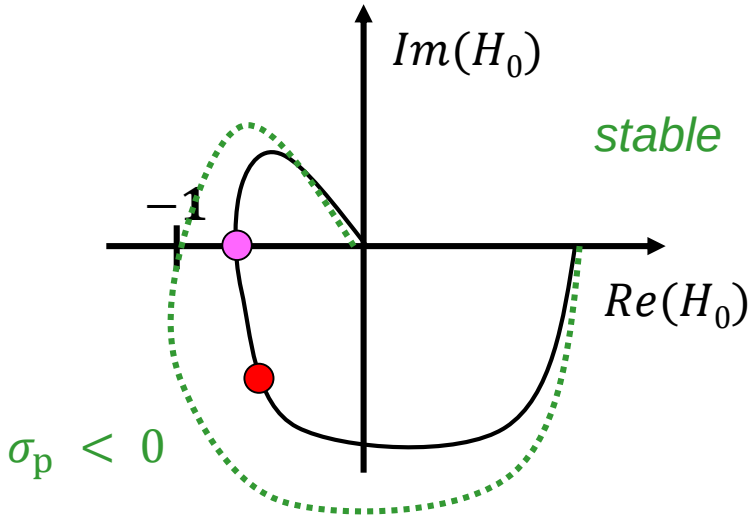
Stability Analysis Based on Nyquist Plot

$1 + H_0(s) = 0$ characteristic equation

$\Leftrightarrow \quad H_0(s) = -1$

Consider -1 in the Nyquist plot &

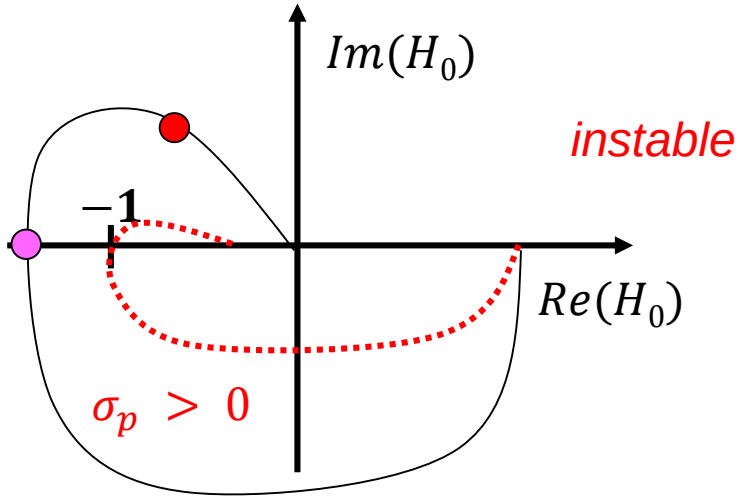
check if Nyquist curve through -1 has $\sigma_p < 0$ or $\sigma_p > 0$



Nyquist stability criteria

Check if Nyquist curve for $s = j\omega$, i.e., $\sigma_p = 0$, has

- an amplitude < 1 @ -180° phase shift
  ↔ Nyquist curve through -1 has $\sigma_p < 0$
- a phase shift < -180° @ amplitude = 1
  ↔ amplitude will be < 1 @ -180°
  ↔ Nyquist curve through -1 has $\sigma_p < 0$



Bode Plot – Nyquist Stability Criteria

$$\frac{v_{\mathrm{OUT}}}{v_{\mathrm{IN}}} = \frac{A(s)}{1 + F(s)A(s)}$$

$F(s)A(s) = -1$?

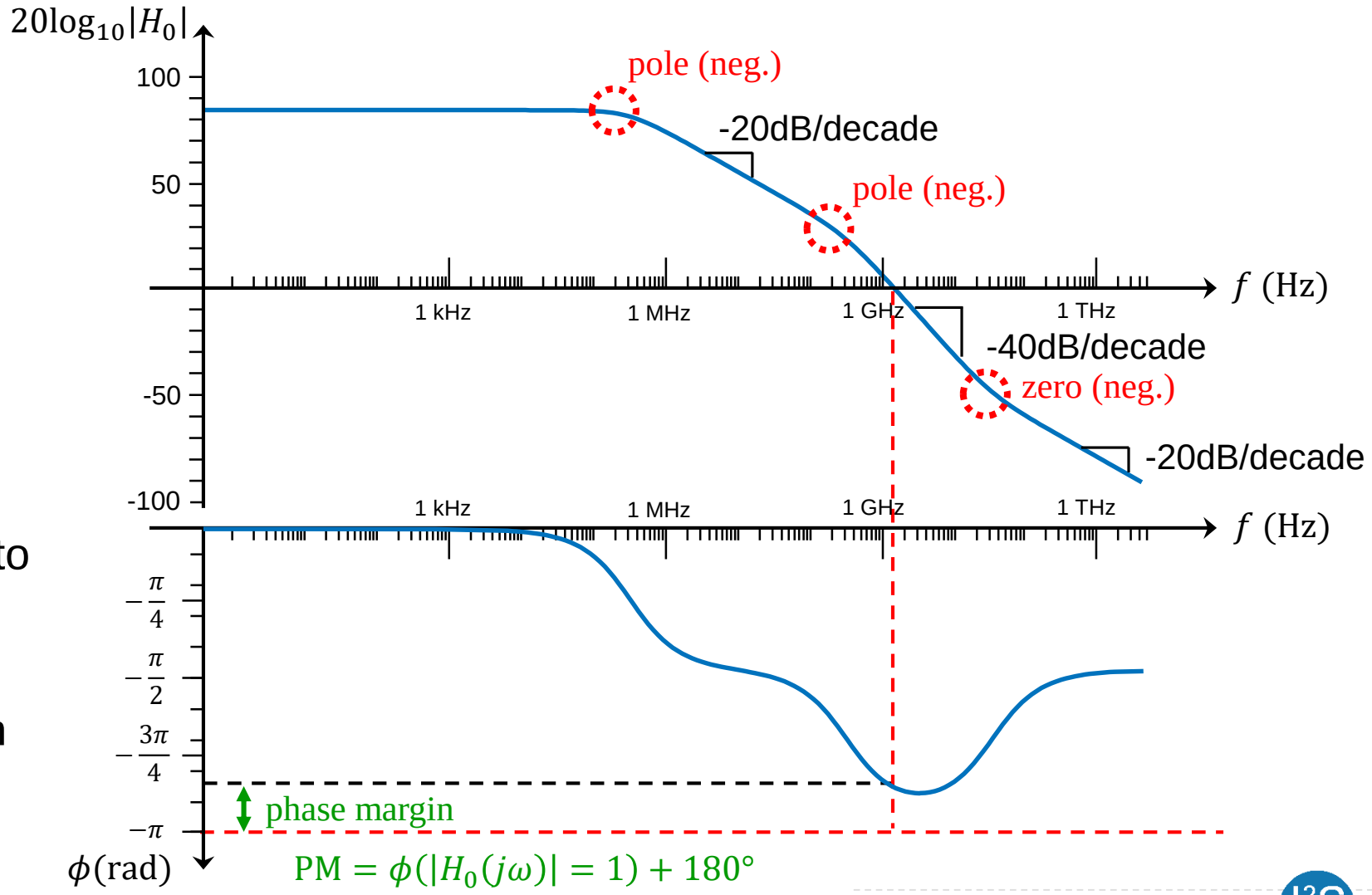
magnitude = 1 (0 dB)?
phase = -180°?

phase margin

If $|H_0(j\omega)| = 1$, i.e., 0 dB, how much phase margin do we have to -180°?

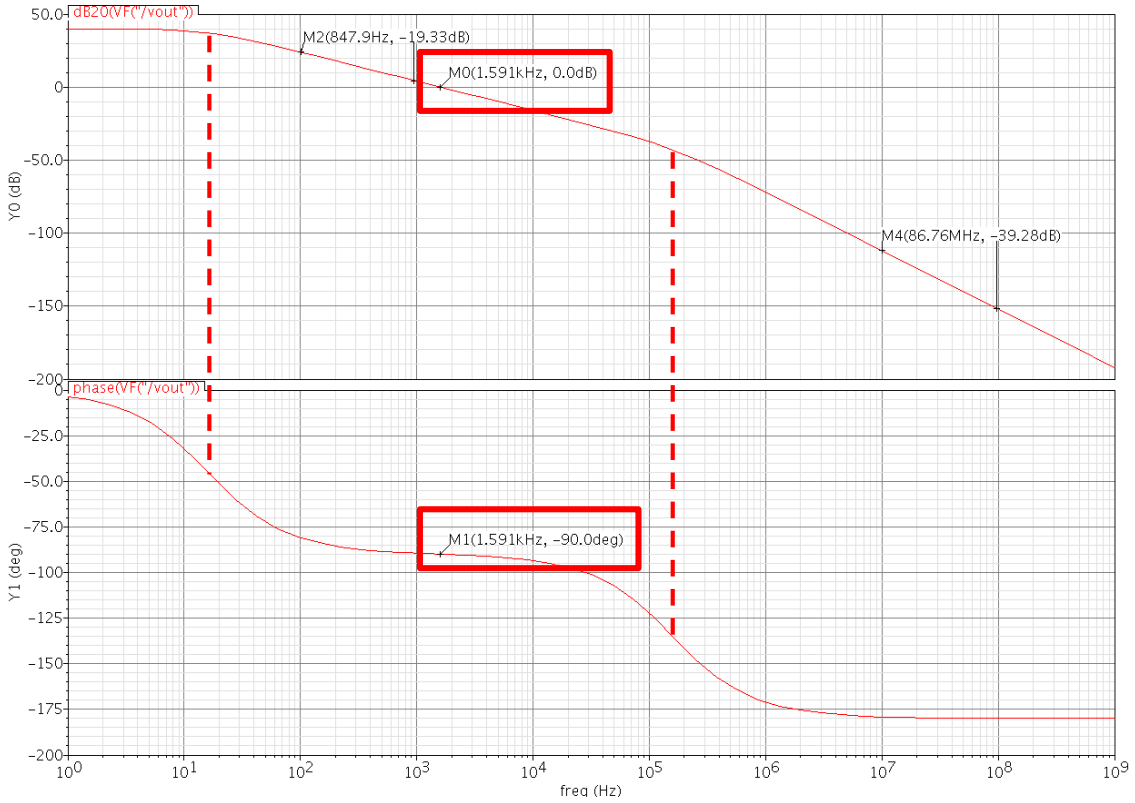
gain margin

If $\phi\big(H_0(j\omega)\big) = 180°$, how much gain margin do we have to $|H_0(j\omega)| = 1$ (0 dB)?

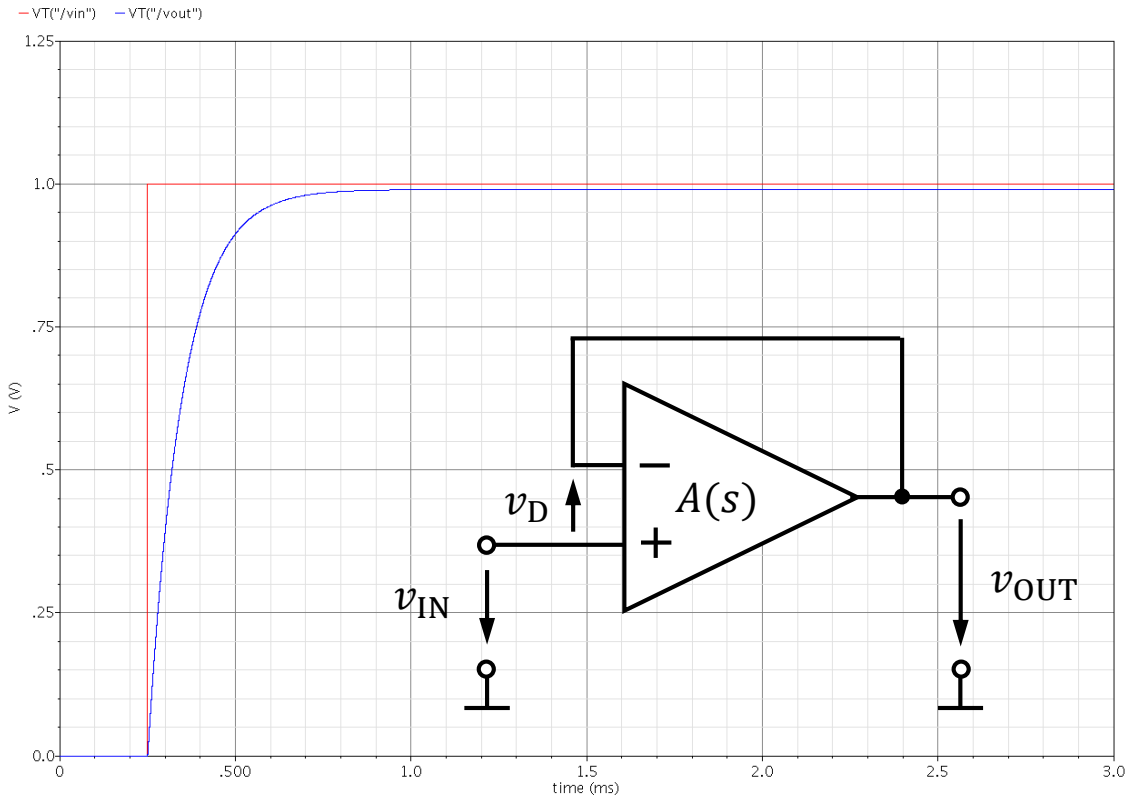


Small-signal Settling Behavior – PM = 90°

Bode plot $H_0(\boldsymbol{j2\pi f})$

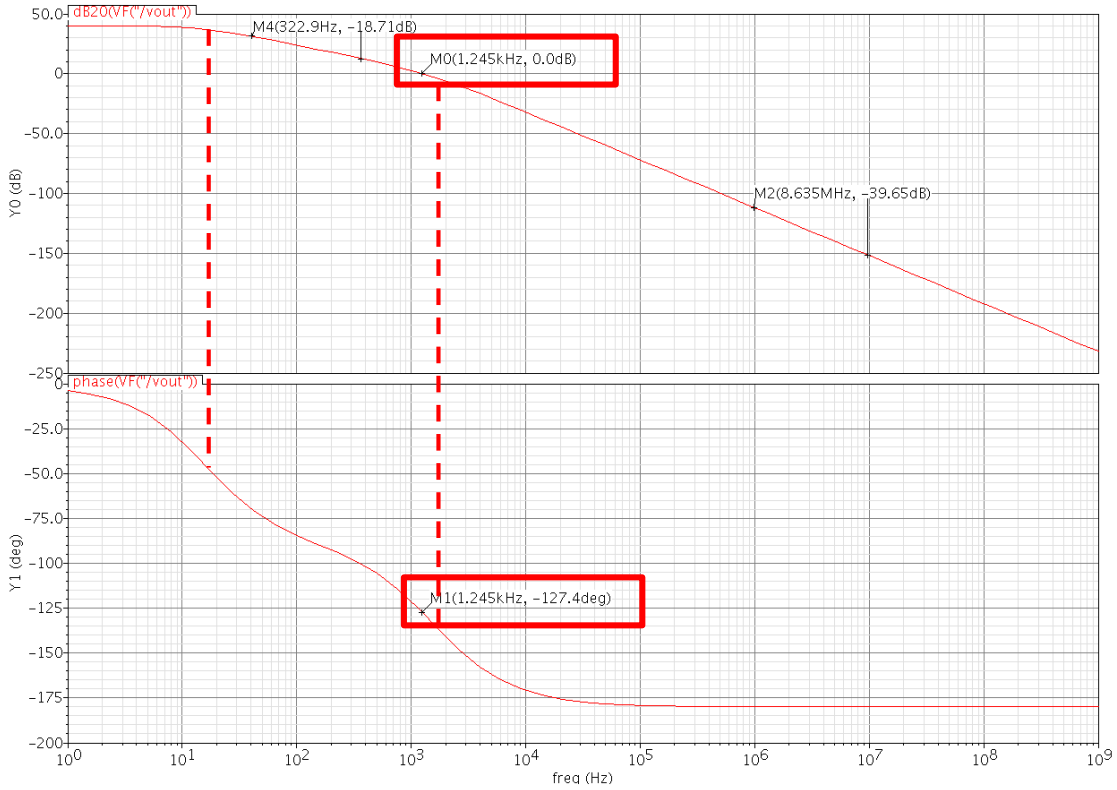


transient step response

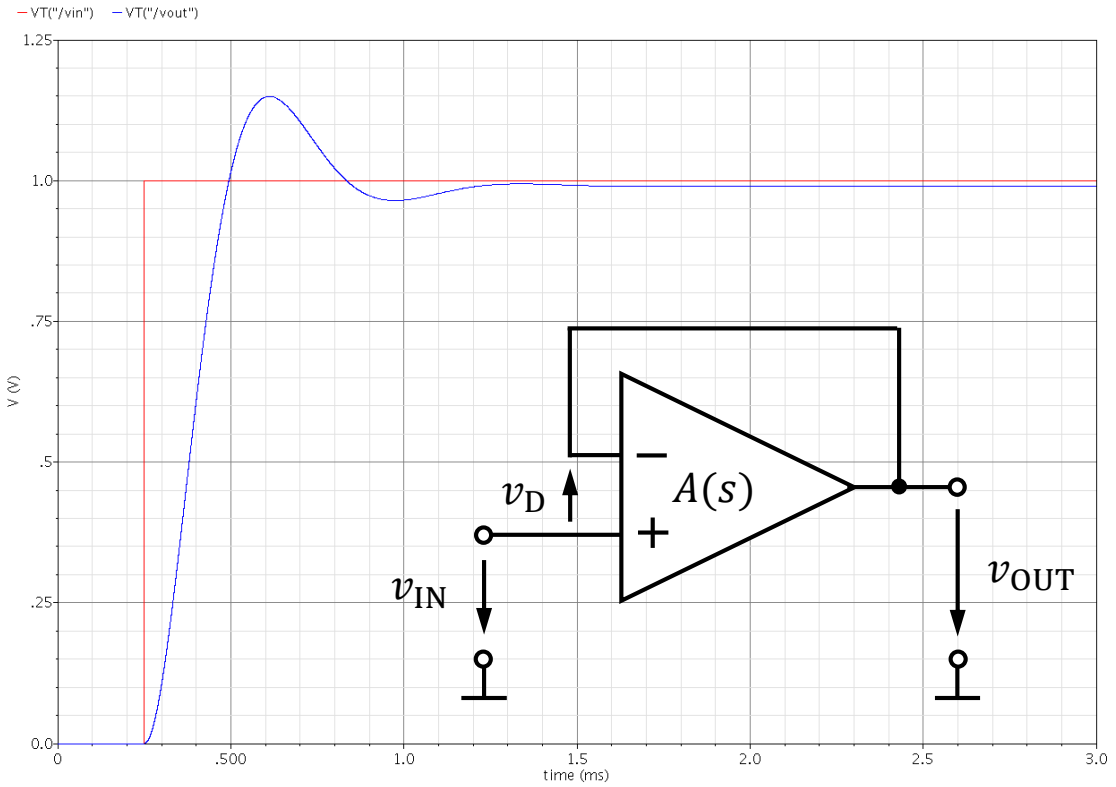


Small-signal Settling Behavior – PM = 53°

Bode plot $H_0(\boldsymbol{j2\pi f})$

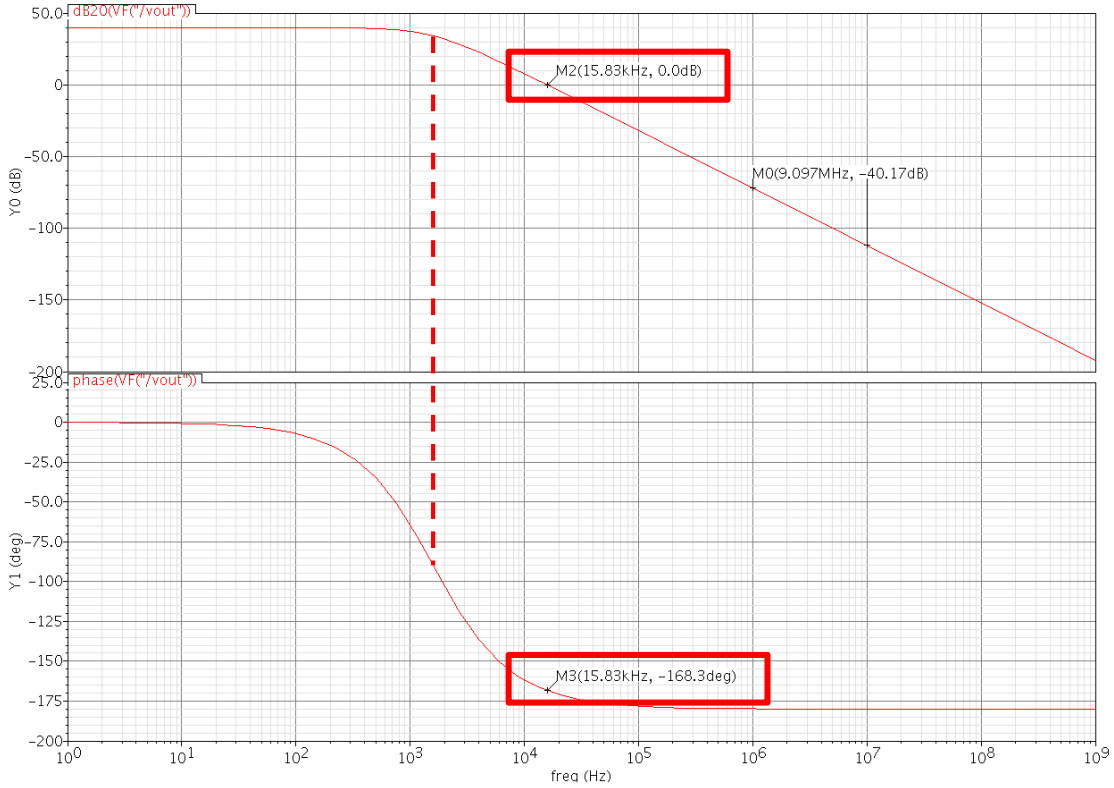


transient step response

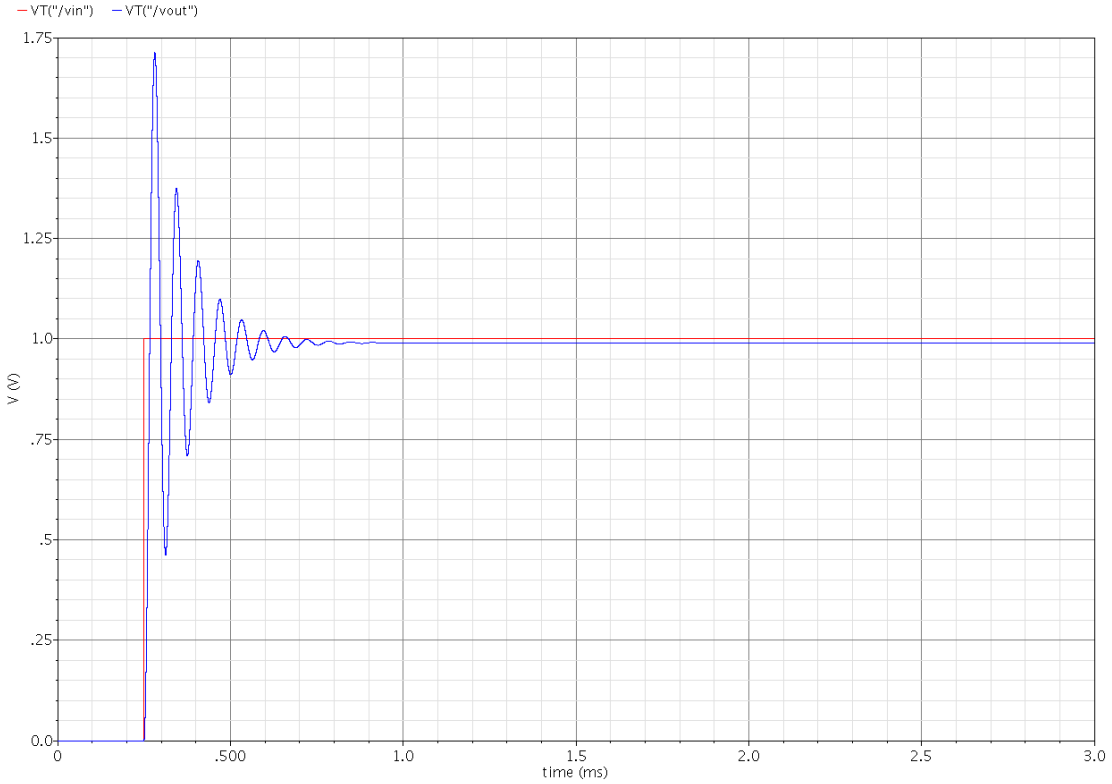


Small-signal Settling Behavior – PM = 12°

Bode plot $H_0(j2\pi f)$



transient step response



Bode Plot – Summary III

- Based on the Bode plot of the loop transfer function, the stability of a closed-loop feedback system is typically analyzed by deriving the phase margin.
- The lower the phase margin, the more ringing in the small-signal settling behavior.

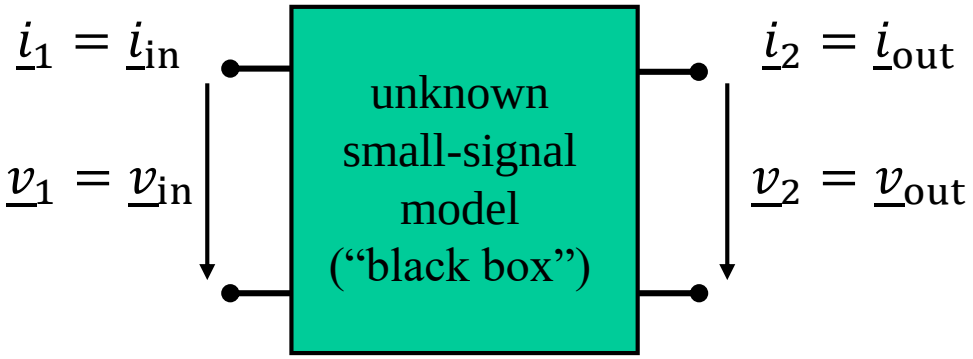
Literature

- Razavi, chapters 10.1-10.3, page 345-355

Overview

- Control Theory & System Characterization – A Summary
- **Two-port Model of the MOS Transistor**
- CS Amplifier – Frequency Response
- Miller-Effect

Recap – Two-port Parameters (Y-Parameters)



The "black box" is characterized by the characteristics of the currents and voltages at the input & output
→ complex admittance parameters

$$\underline{y}_{11}, \underline{y}_{12}, \underline{y}_{21}, \underline{y}_{22}$$

complex admittance $\underline{Y} = \frac{1}{\underline{Z}}$ complex impedance

$$(\underline{G}_{\text{m}} =)\,\underline{y}_{21} = \left.\frac{\underline{i}_{\text{out}}}{\underline{v}_{\text{in}}}\right|_{\underline{v}_{\text{out}} = 0\ \text{V}}$$

$$(\underline{g}_{\text{out}} =)\,\underline{y}_{22} = \left.\frac{\underline{i}_{\text{out}}}{\underline{v}_{\text{out}}}\right|_{\underline{v}_{\text{in}} = 0\ \text{V}}$$

input admittance

reverse transfer admittance

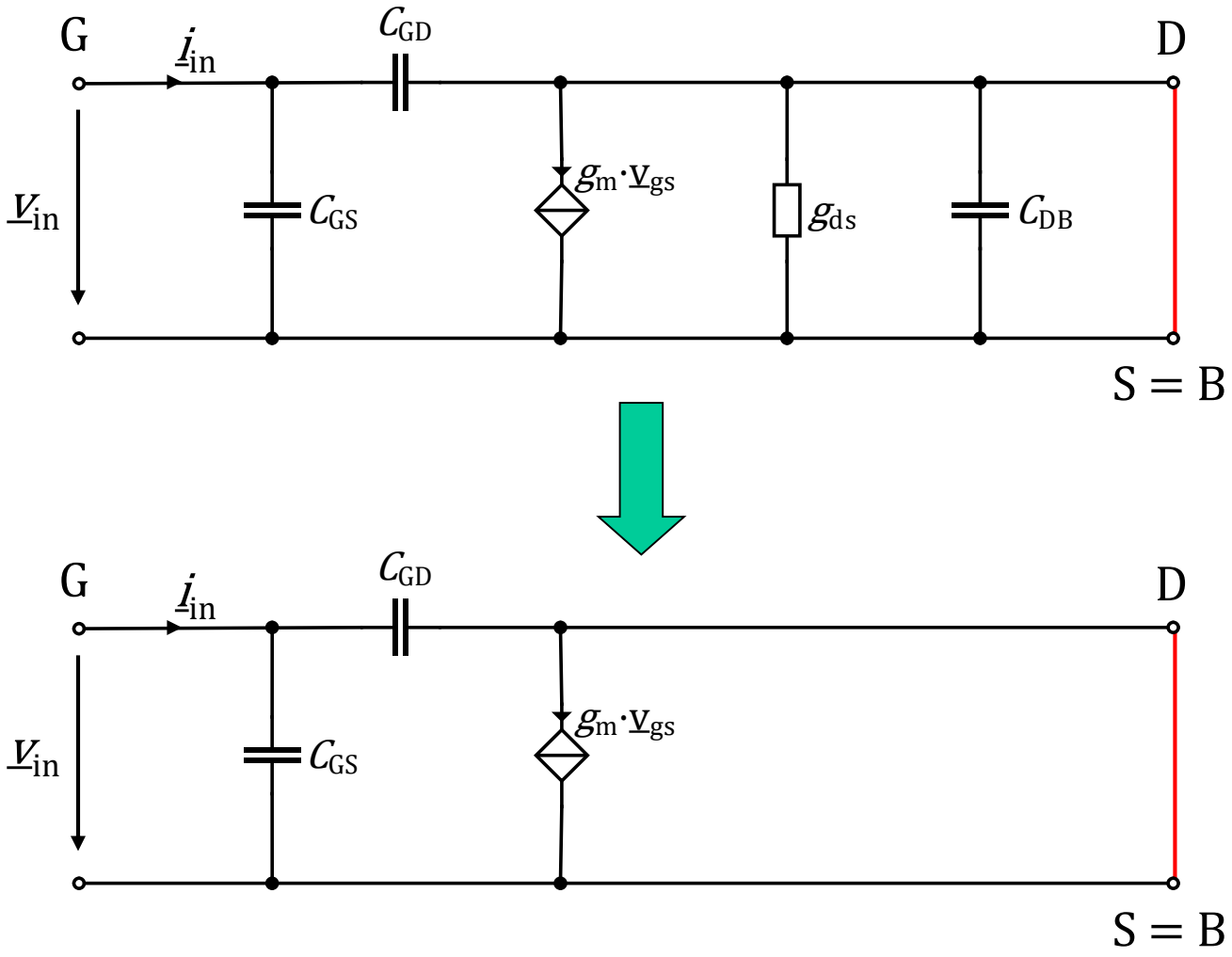
$$\underline{i}_{\text{in}} = \underline{y}_{11}\underline{v}_{\text{in}} + \underline{y}_{12}\underline{v}_{\text{out}}$$

$$\underline{i}_{\text{out}} = \underline{y}_{21}\underline{v}_{\text{in}} + \underline{y}_{22}\underline{v}_{\text{out}}$$

transconductance

output admittance

MOS-Transistor – Input Admittanz (y_{11})



$$\underline{y}_{11} = \left.\frac{\underline{i}_{in}}{\underline{v}_{in}}\right|_{\underline{v}_{out} = 0\,V}$$

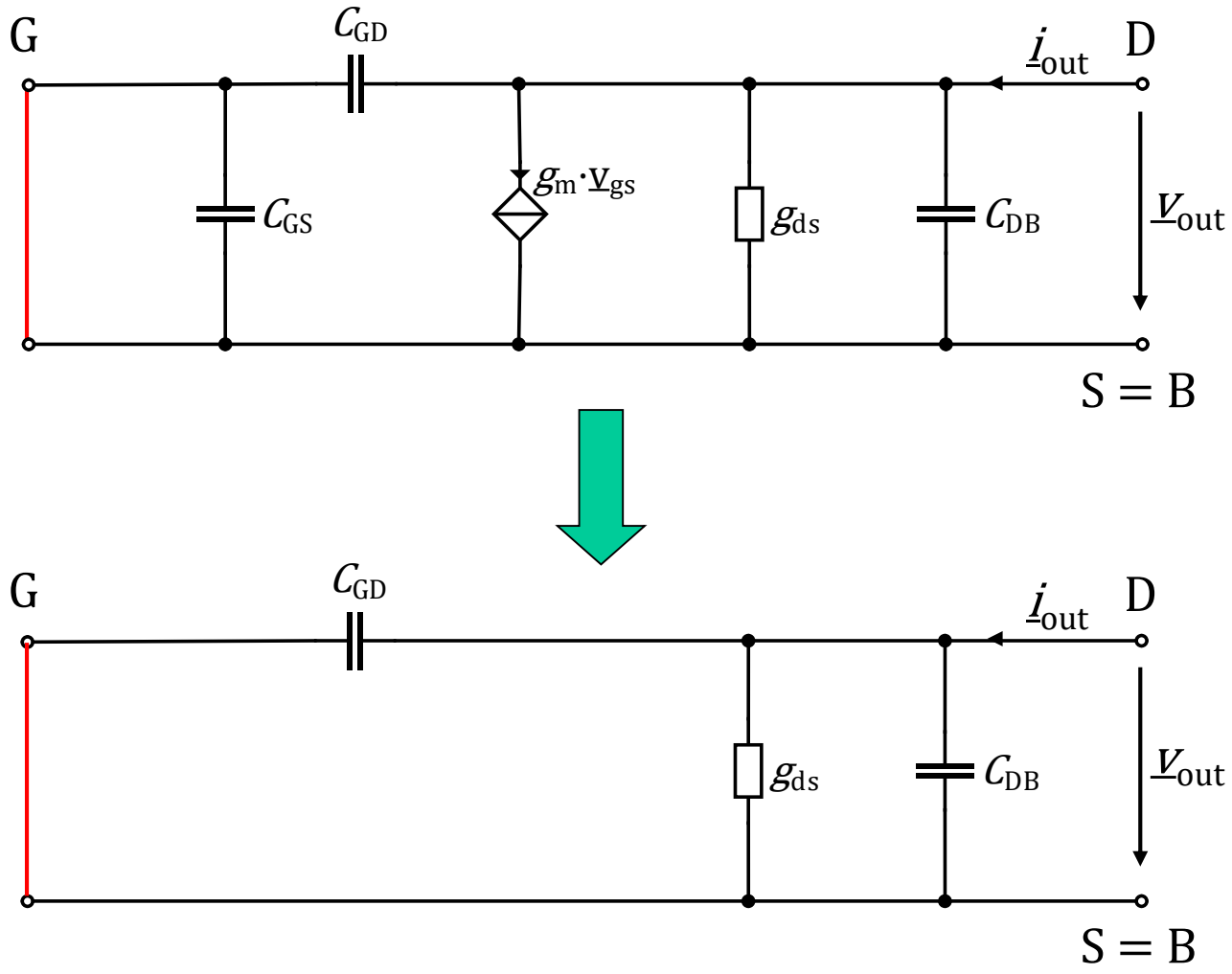
$$= \underline{y}_{C_{GS}} + \underline{y}_{C_{GD}}$$

$$= s(C_{GS} + C_{GD})$$

$$\underline{z}_{in} = \frac{1}{\underline{y}_{11}} = \frac{1}{s(C_{GS} + C_{GD})}$$

frequency response
→ replace s by
complex frequency $j\omega$

MOS-Transistor – Output Admittanz (y_{22})



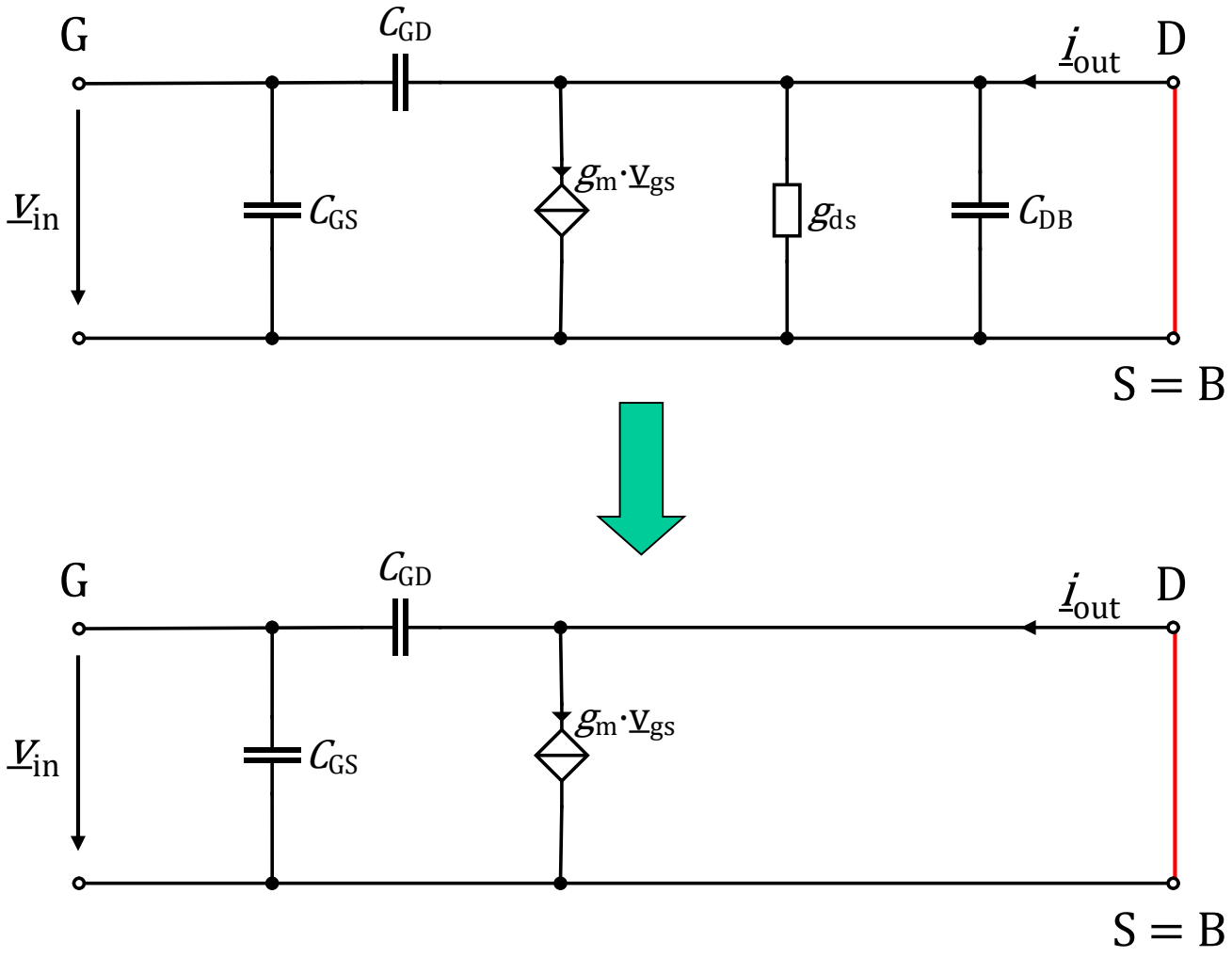
$$\underline{y}_{22} = \left.\frac{\underline{i}_{out}}{\underline{v}_{out}}\right|_{\underline{v}_{in}=0\,V}$$

$$= g_{ds} + \underline{y}_{C_{GD}} + \underline{y}_{C_{DB}}$$

$$= g_{ds} + (C_{GD} + C_{DB})s$$

$$\underline{z}_{out} = \frac{1}{\underline{y}_{22}} = \frac{r_{ds}}{1 + r_{ds}(C_{GD} + C_{DB})s}$$

MOS-Transistor – Transconductance ($\underline{y}_{21}$)

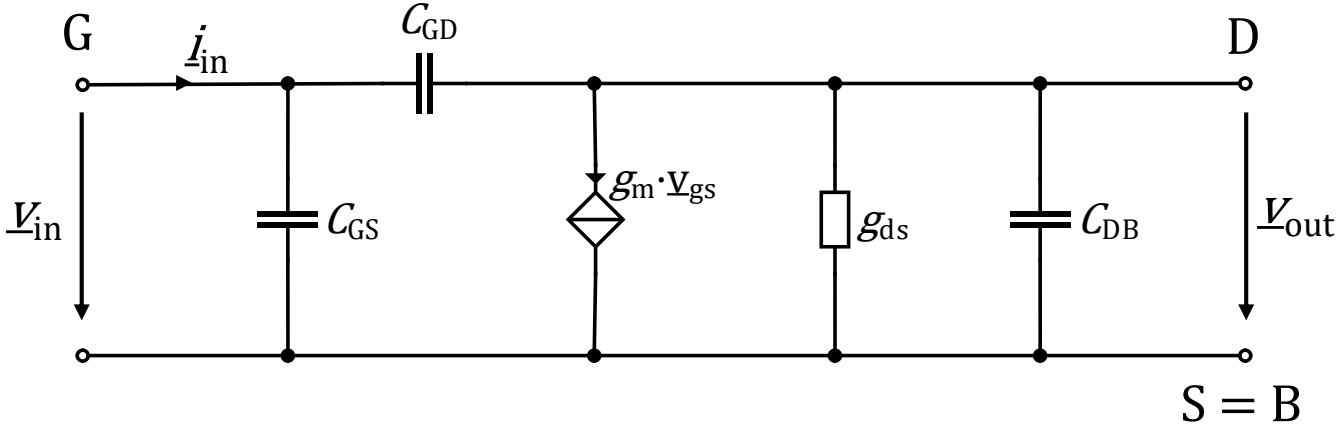


$$\underline{y}_{21} = \left.\frac{\underline{i}_{out}}{\underline{v}_{in}}\right|_{\underline{v}_{out} = 0\,V} = \frac{g_m \underline{v}_{gs} - sC_{GD}\underline{v}_{C_{GD}}}{v_{in}}$$

$$\underline{G}_m = \frac{1}{\underline{y}_{21}} = g_m - sC_{GD}$$

$$\underline{v}_{in} = \underline{v}_{gs} = \underline{v}_{C_{GS}} = \underline{v}_{C_{GD}}$$

MOSFET – Frequency-dependent Small-signal Gain



$$G_m = \underline{Y}_{21} = g_m - sC_{GD} \qquad \underline{r}_{out} = \frac{1}{\underline{Y}_{22}} = \frac{r_{ds}}{1 + s(C_{GD} + C_{DB})r_{ds}}$$

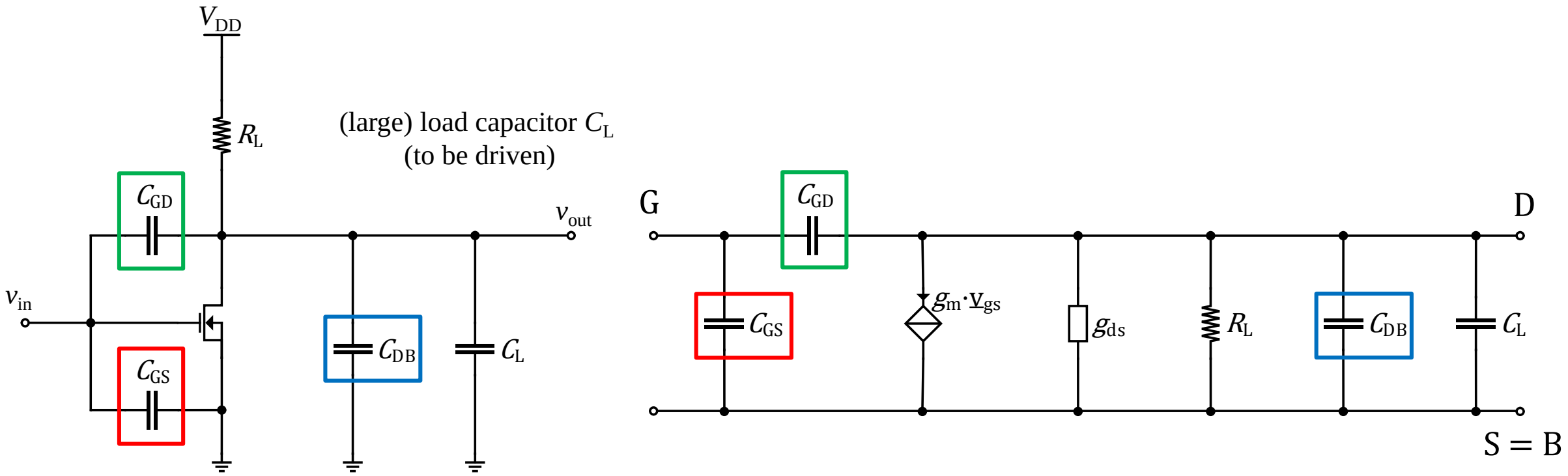
$$\boxed{\underline{A}_V = -(g_m - sC_{GD})\frac{r_{ds}}{1 + s(C_{GD} + C_{DB})r_{ds}}}$$

$$\omega \to 0\ \text{rad/s}: A_V = -g_m r_{ds}$$

Overview

- Control Theory & System Characterization – A Summary
- Two-port Model of the MOS Transistor
- **CS Amplifier – Frequency Response**
- Miller-Effect

CS Amplifier – Frequency-dependent Small-signal Gain



$$A_V = -G_m r_{out}$$

$$= -(g_m - sC_{GD}) \frac{r_{ds} || R_L}{1 + s(C_{GD} + C_{DB} + C_L)(r_{ds} || R_L)}$$

$$\omega \to 0 \text{ rad/s}: \; A_V = -g_m (r_{ds} || R_L)$$

CS Amplifier – Frequency Response

$$A_\mathrm{V} = -(g_\mathrm{m} - sC_\mathrm{GD})\left(\frac{r_\mathrm{ds}||R_\mathrm{L}}{1 + s(C_\mathrm{GD} + C_\mathrm{DB} + C_\mathrm{L})(r_\mathrm{ds}||R_\mathrm{L})}\right)$$

pole(s): Solve $denominator(A_\mathrm{V}) = 0$!

$$\Rightarrow s_\mathrm{p} = -\frac{1}{(r_\mathrm{ds}||R_\mathrm{L})(C_\mathrm{GD} + C_\mathrm{DB} + C_\mathrm{L})} \approx -\frac{1}{R_\mathrm{L}C_\mathrm{L}}$$

$$|s_\mathrm{p}| = |j\omega_\mathrm{p}| = |\omega_\mathrm{p}| \approx \frac{1}{R_\mathrm{L}C_\mathrm{L}} = 10^7 \ \frac{\mathrm{rad}}{\mathrm{s}} \Rightarrow |f_\mathrm{p}| = \frac{|\omega_\mathrm{p}|}{2\pi} = 1.59 \ \mathrm{MHz}$$

zero(s): Solve $numerator(A_\mathrm{V}) = 0$!

$$\Rightarrow s_\mathrm{z} = +\frac{g_\mathrm{m}}{C_\mathrm{GD}}$$

$$|s_\mathrm{z}| = |j\omega_\mathrm{z}| = |\omega_\mathrm{z}| = \frac{g_\mathrm{m}}{C_\mathrm{GD}} = 18.6 \cdot 10^9 \ \frac{\mathrm{rad}}{\mathrm{s}} \Rightarrow |f_\mathrm{p}| = \frac{|\omega_\mathrm{z}|}{2\pi} = 2.96 \ \mathrm{GHz}$$

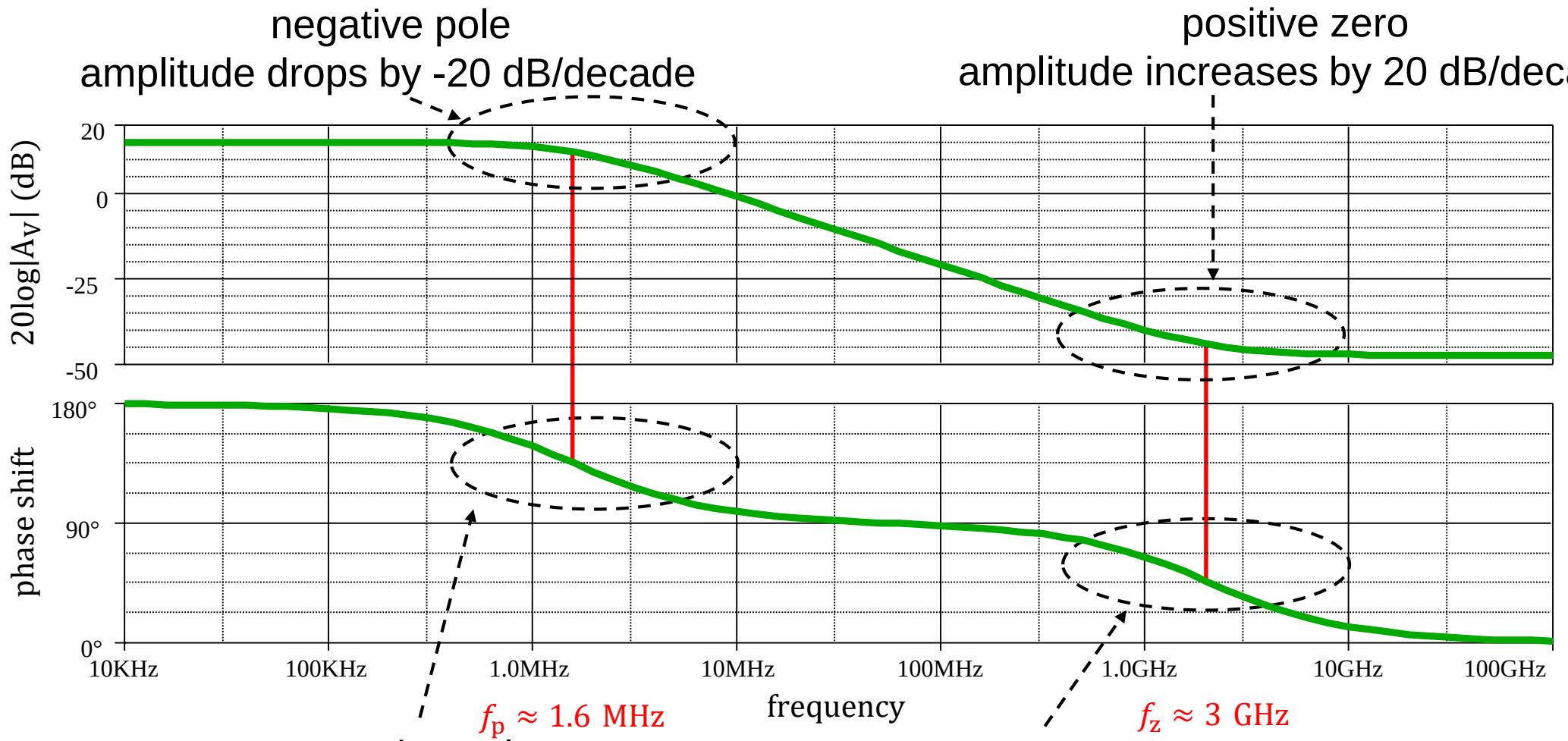
parameters:

$$\frac{W}{L} = \frac{10 \ \mu m}{10 \ \mu m}$$
$$V_\mathrm{GS} = 0.35 \ \mathrm{V}$$
$$R_\mathrm{L} = 100 \ \mathrm{k\Omega}$$
$$C_\mathrm{L} = 1 \ \mathrm{pF}$$

simulation results:

$$g_\mathrm{m} = 57.7 \ \mu\mathrm{S}$$
$$r_\mathrm{ds} = 10 \ \mathrm{M\Omega}$$
$$C_\mathrm{GD} = 3.1 \ \mathrm{fF}$$
$$C_\mathrm{DB} = 0.07 \ \mathrm{fF}$$

CS Amplifier – Bode Plot

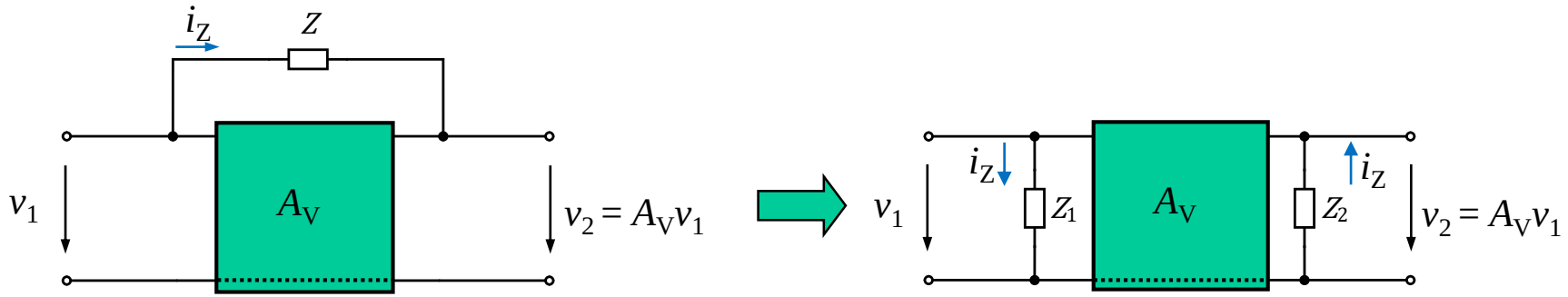


Overview

- Control Theory & System Characterization – A Summary
- Two-port Model of the MOS Transistor
- CS Amplifier – Frequency Response
- **Miller-Effect**

Miller-Theorem / Miller-Effect

An impedance $\underline{Z}$ connected between input and output of an amplifier can be replaced by an input and output referred impedance $\underline{Z_1}$ and $\underline{Z_2}$, respectively.



$$i_Z = \frac{v_1 - v_2}{Z} = \frac{v_1 - A_V v_1}{Z} = \frac{v_1(1 - A_V)}{Z} \quad (1)$$

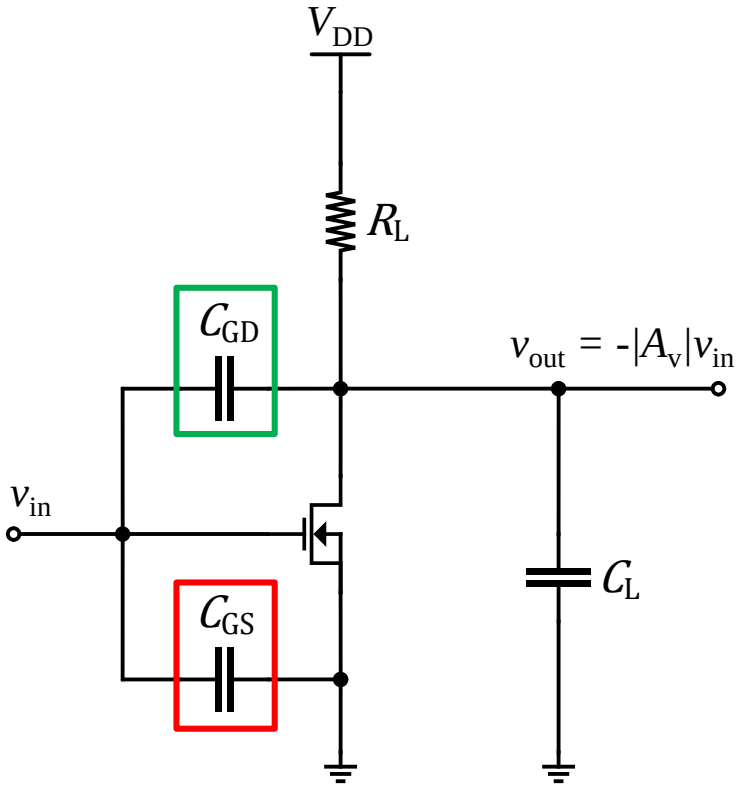
$$i_Z = \frac{v_1}{Z_1} \quad (2)$$

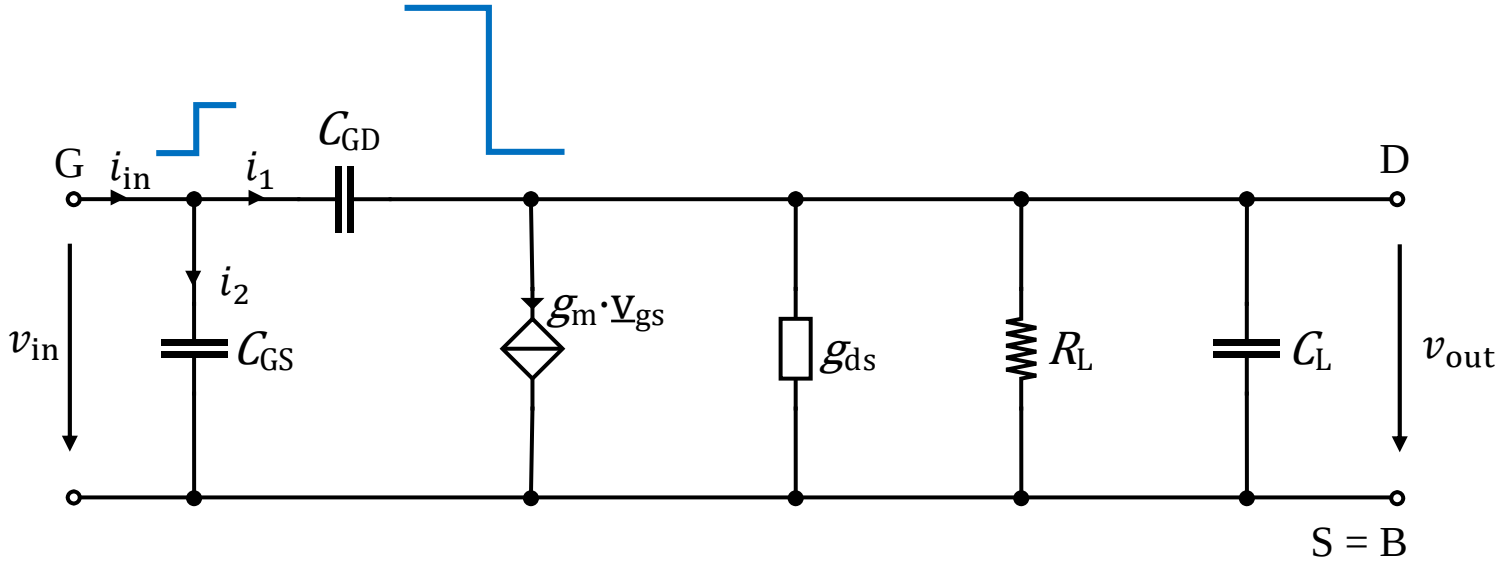
$$(1) = (2) \Rightarrow \boxed{Z_1 = \frac{Z}{1 - A_V}}$$

homework

Prove that $Z_2 = \frac{1}{1 - \frac{1}{A_V}} Z \approx Z$!

CS Amplifier – "Pseudo" Input Impedance

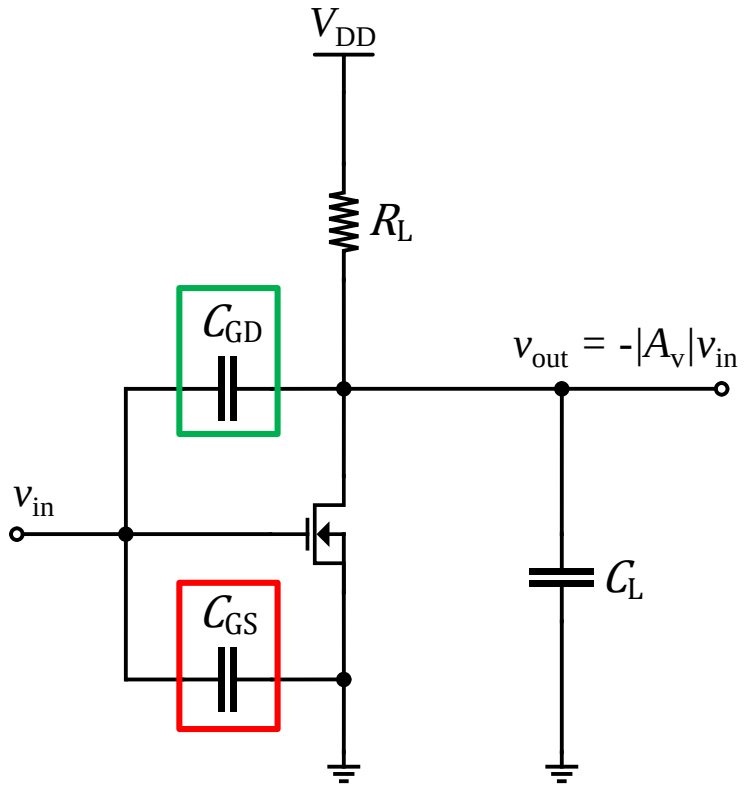




$$z_{in} = \frac{v_{in}}{i_{in}}\bigg|_{v_{out} \neq 0\,V} = \frac{v_{in}}{sC_{GS}v_{in} + sC_{GD}(v_{in} - v_{out})} = \frac{1}{sC_{GS} + s(1 + |A_v|)C_{GD}} \approx \frac{1}{s(C_{GS} + |A_v|C_{GD})}$$

"Miller-effect": C_{GD} appears as "large" capacitor at the amplifier input!

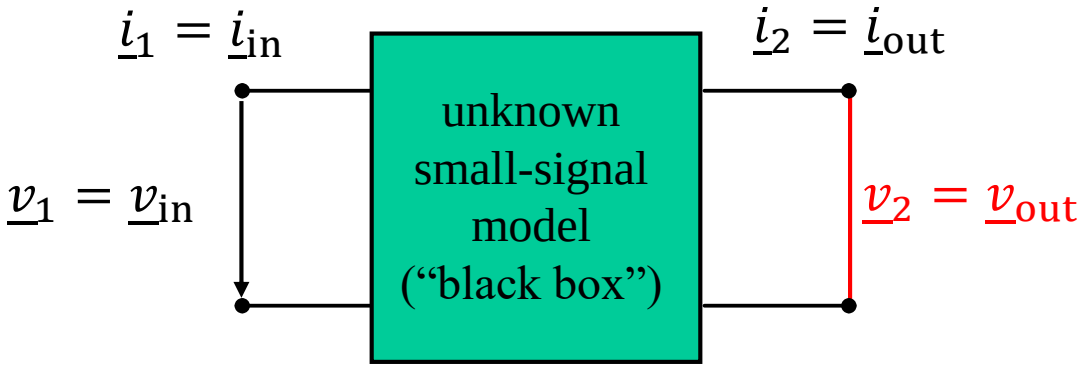
CS-Amplifier – Miller-Effect

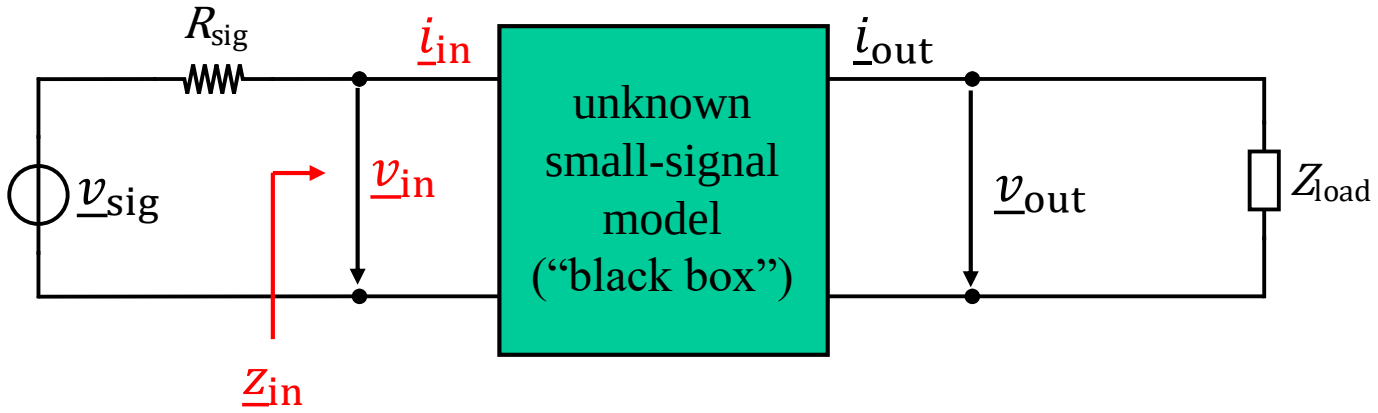


$$C_{GD} = 0.1\text{pF} \rightarrow Z = 1/sC_{GD}$$
$$A_V = -100$$

$$Z_1 = \frac{Z}{1 - A_V} \rightarrow C_M = (1 - A_V)C_{GD} = 10.1\ \text{pF}$$

Don‘t Mix Up!





$$\underline{i}_{in} = \underline{y}_{11}\underline{v}_{in} + \underline{y}_{12}\underline{v}_{out}$$

$$\underline{i}_{out} = \underline{y}_{21}\underline{v}_{in} + \underline{y}_{22}\underline{v}_{out}$$

short-circuit input impedance
=
parameter of the two-port model

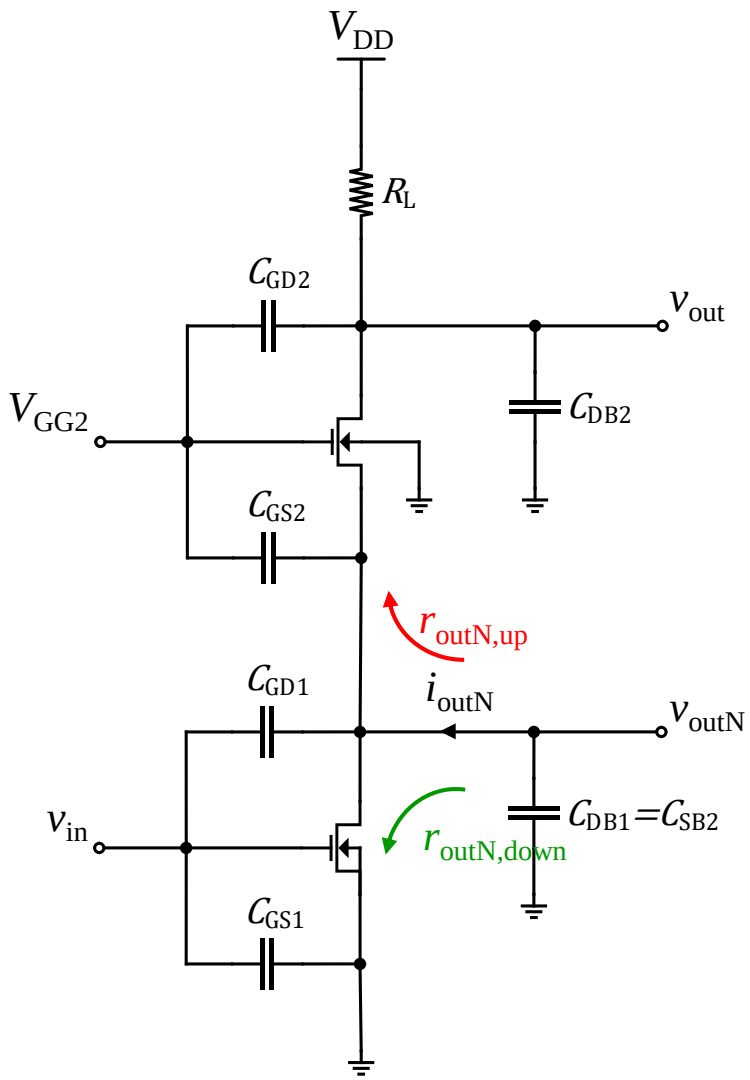
$$\underline{z}_{in} = \frac{1}{\underline{y}_{11}} = \frac{\underline{v}_{in}}{\underline{i}_{in}}\bigg|_{\underline{v}_{out} = 0\ \mathrm{V}}$$

input impedance
=
impedance, the signal source must drive

$$\underline{z}_{in} = \frac{\underline{v}_{in}}{\underline{i}_{in}}$$

(either without = open-circuit or with load)

CS Cascode Amplifier – Miller-Effect



$$C_{IN} = C_{GS1} + (1 + |A_{V1}|)C_{GD1}$$

$A_{V1} = ?$ => gain from input to N

$$A_{V1} = -g_{m1} r_{outN}$$

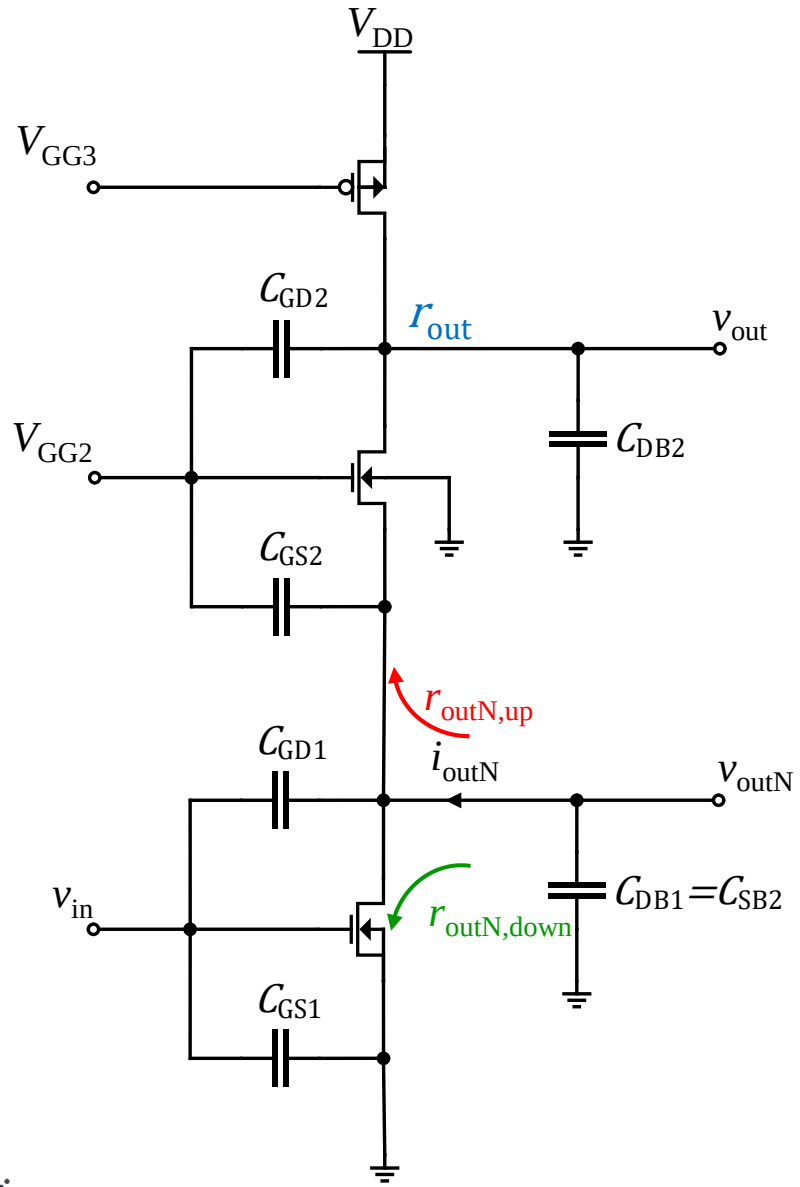
$$= -g_{m1} \left(r_{outN,up} \,\|\, r_{outN,down} \right)$$

$$r_{outN,up} = \frac{r_{ds2} + R_L}{1 + (g_{m2} + g_{mb2}) r_{ds2}}$$

exercise at home

$$r_{outN,down} = r_{ds1}$$

CS Cascode Amplifier with Current Source Load – Miller-Effect



$$r_{outN,up} = \frac{r_{ds2} + R_L}{1 + (g_{m2} + g_{mb2})r_{ds2}} \approx \frac{r_{ds2} + r_{ds3}}{g_{m2} r_{ds2}} \approx \frac{2}{g_{m2}}$$

$$r_{outN,down} = r_{ds1}$$

$$r_{outN} \approx \frac{2}{g_{m2}} \Rightarrow A_V \approx -2\frac{g_{m1}}{g_{m2}}$$

$$\boxed{C_M \approx \left(1 + 2\frac{g_{m1}}{g_{m2}}\right) C_{GD}}$$

low gain from input to node N
→ minor Miller-effect

gain of complete stage

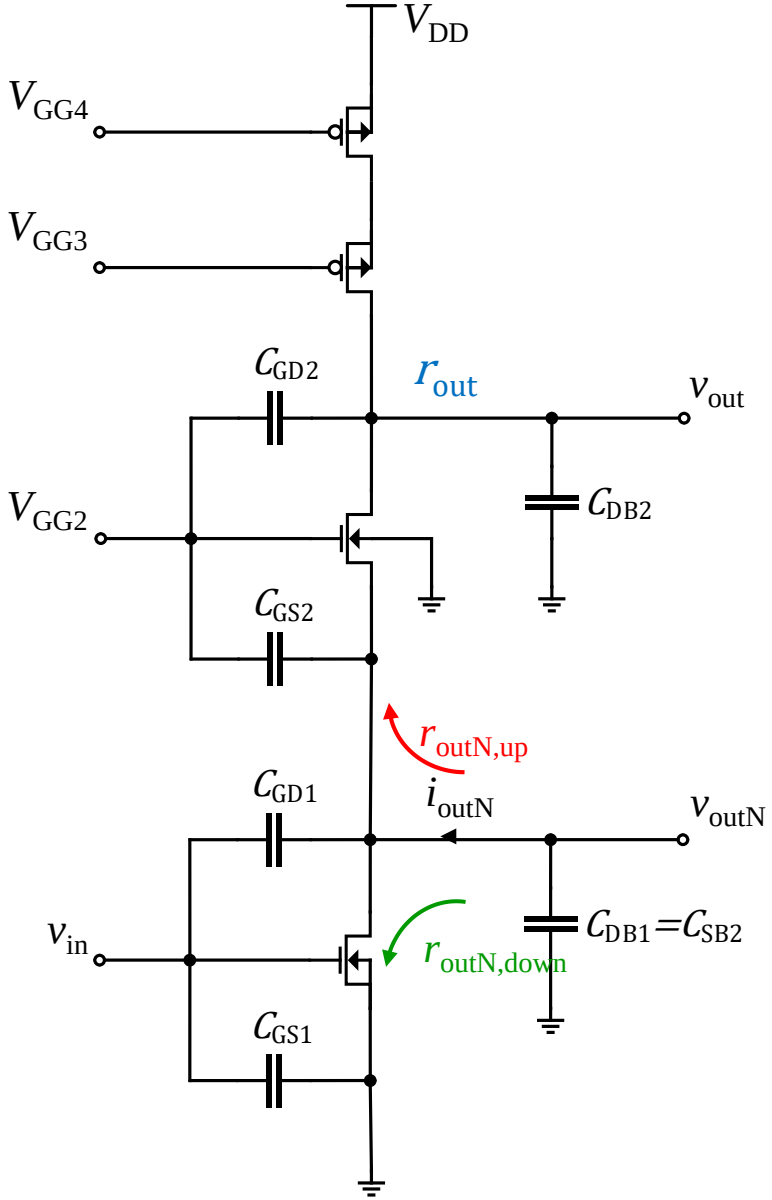
$$A_V = -g_{m1} r_{out}$$

$$r_{out} \approx (g_{m2} r_{ds2} r_{ds1}) || r_{ds3}$$

$$\boxed{A_V \approx -g_{m1} r_{ds3}}$$

medium overall gain from input to output

Full Cascode CS Amplifier – Miller-Effect



$$r_{outN,up} = \frac{r_{ds2} + R_L}{1 + (g_{m2} + g_{mb2})r_{ds2}} \approx \frac{r_{ds2} + g_{m3}r_{ds3}r_{ds4}}{g_{m2}r_{ds2}} \approx r_{ds4}$$

$$r_{outN,down} = r_{ds1}$$

$$r_{outN} \approx r_{ds1} || r_{ds4} \Rightarrow A_V \approx g_{m1}(r_{ds1} || r_{ds4})$$

$$\boxed{C_M \approx g_{m1}(r_{ds1} || r_{ds4})C_{GD}}$$

medium gain from input to node N
→ medium Miller-effect

gain of complete stage

$$A_V = -g_{m1}r_{out}$$

$$r_{out} \approx (g_{m2}r_{ds2}r_{ds1}) || (g_{m3}r_{ds3}r_{ds4})$$

$$\boxed{A_V \approx -\frac{1}{2}g_{m1}g_{m2}r_{ds2}r_{ds1}}$$

high overall gain from input to output

Summary

amplifier	CS amplifier (transistor load r_{ds})	single-side cascode (transistor load r_{ds})	full cascode
output resistance r_{out}	$r_{ds}/2$	r_{ds}	$1/2 g_{m3} r_{ds3} r_{ds4}$
output resistance r_{outN} @ node N		$\approx \frac{1}{g_{m2}}$	$\approx r_{ds1} \| r_{ds4}$
“pseudo” input capacitance due to Miller-effect	$(1 + \|A_V\|) C_{GD}$	$\approx \frac{g_{m1}}{g_{m2}} C_{GD} \approx C_{GD}$	$\approx g_{m1} (r_{ds1} \| r_{ds4}) C_{GD}$
overall gain A_V	$-g_{m1} r_{ds}/2$	$\approx -g_{m1} r_{ds}$	$\approx -\frac{1}{2} (g_m r_{ds})^2$
remarks	medium gain Miller-effect	medium gain high decoupling	high gain medium decoupling

Literature

- Razavi, chapters 6.2, page 172-178
- Razavi, chapters 6.5, page 185-187